Biochemistry CUME

April 25, 1998

In noncyclic photosynthetic electron transport, electrons from water are used to reduce NADP to NADPH. This electron transport chain involves both Photosystem I and II and a quinol oxidizing cytochrome b6/f complex in addition to other electron carriers. Electron transfer is coupled to formation of a protonmotive force such that an amount of ATP is formed approximately equal to the amount of NADP reduced. Additional ATP is generated via cyclic electron transport around Photosystem I in a process also known to involve the cytochrome b6/f complex.

Ferredoxin:NADP reductase (FNR) catalyzes the terminal step of photosynthetic electron transport, using electrons from ferredoxin to reduce NADP to NADPH. Ferredoxin receives electrons from the reducing side of Photosystem I. Ferredoxin has also been demonstrated to be an essential electron carrier in cyclic electron transport. Existing evidence is ambiguous as to whether FNR is also involved in cyclic electron transport or whether electrons from ferrredoxin are transferred to something else (bypassing FNR) in the case of cyclic electron transport. This is not an easy problem and one which has remained unresolved in the photosynthesis research community for nearly 30 years!

Propose at least two experimenal approaches which you feel would definatively determine whether or not FNR is involved in cyclic electron transfer. Be as thorough and specific as time and knowledge allow but also think creatively.

You will be graded based on

A. the thoroughness of your description and justification of the experimenatal approaches chosen (50 points),

B. a description as to how the results of each would be interpreted (what observations would indicate definitively that FNR is or is not involved in cyclic electron transfer?) and whether there are alternative explanations of your results (the main reason why the question has been unresolved for so long) (50 points)